The Carnival Caper

Whodunnit? · Circles

Name: _____

It was the last night of the Sunny Days Carnival, and the biggest prize, the **Golden Ticket Trophy**, shone on its stand. When the Ferris wheel lit up for the finale, the trophy stand was bare!

Six people were at the carnival, each in a different spot, and there are six sneaky ways to carry a trophy off the fairgrounds. You are the lead detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Ringo Ride Operator	☐ Ferris Wheel	☐ Hidden in balloons
☐ Ms. Taffy Candy Vendor	☐ Cotton Candy Stand	☐ Stuffed in a prize toy
☐ Mr. Dunk Game Booth Owner	☐ Dunk Tank	☐ Carried in a candy cone
☐ Penny Ticket Seller	☐ Ticket Booth	☐ Hauled on a ride cart
☐ Clara Face Painter	☐ Hall of Mirrors	☐ Hidden in the popcorn machine
☐ Big Lou Balloon Seller	☐ Bumper Cars	☐ Dropped in a bumper car

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Carnival Caper · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Carnival Caper

Find the circumference of a circle with radius 5. Leave your answer in terms of π.

A. 25π	Big Lou was waiting in line for the Ferris Wheel with a bunch of balloons. The operator remembers him.
B. 10π	Ringo was running the Ferris Wheel. His ride log has every loading time.
C. 5π	Ringo was tightening bolts near the Hall of Mirrors. Two kids asked him for directions.
D. 20π	Every balloon bunch was popped and checked. Nothing inside but air.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Carnival Caper

Find the area of a circle with radius 6. Leave your answer in terms of π.

A. 36π	The prize toys were all checked at the exit. None held the trophy.
B. 144π	The balloons by the gate were all squeezed. No bulge, no trophy.
C. 12π	Big Lou was buying a cotton candy at the Cotton Candy Stand. Ms. Taffy handed it over.
D. 6π	Ms. Taffy was straightening the mirrors in the Hall of Mirrors. A group of kids counted her reflections.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Carnival Caper

Find the area of a circle with diameter 14. Leave your answer in terms of π.

A. 196π	Big Lou was trying his luck at the Dunk Tank. He got soaked and the crowd cheered.
B. 14π	Mr. Dunk was fixing a latch in the Hall of Mirrors. His toolbox log shows the time.
C. 49π	Ms. Taffy was spinning cotton candy at her stand with a long line of kids.
D. 7π	The balloon cart was unloaded and every balloon examined. All ordinary.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Carnival Caper

A central angle measures 80°. Find the measure of an inscribed angle that intercepts the same arc.

A. 160°	Penny was counting change in the Hall of Mirrors on her break. A guard saw her go in.
B. 80°	Big Lou was asking Penny for a ticket at the Ticket Booth. Her roll shows his purchase.
C. 20°	A balloon bouquet was slit open at the exit. Only string and helium.
D. 40°	The candy cones are all empty paper and sugar. Nothing inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Carnival Caper

An inscribed angle measures 35°. Find the measure of its intercepted arc.

A. 70°	Mr. Dunk was getting dunked in the Dunk Tank for charity. Hundreds watched.
B. 17.5°	The helium tank and every balloon were inspected. Nothing was tucked in any of them.
C. 140°	Clara was painting a star on a kid's cheek in the Hall of Mirrors. The kid still has it.
D. 35°	Big Lou was bumping around the Bumper Cars with his nephew. The ride photo proves it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Carnival Caper

Point P is 13 units from the center O of a circle with radius 5. A tangent from P touches the circle at T. Find PT.

A. 18	Not a single balloon felt heavy. The trophy could not be hidden in one.
B. 12	The ride carts were wheeled back empty and checked.
C. $\sqrt{194}$	Big Lou was riding the Ferris Wheel at the top when the lights flickered. A photo caught him waving.
D. 8	Clara was giving a butterfly to a kid in the Hall of Mirrors. Her brush is still damp.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Carnival Caper

Find the length of an arc with radius 9 and central angle 40°. Leave your answer in terms of π.

A. 18π	Ringo was announcing the next ride from the Hall of Mirrors. The loudspeaker clip is timed.
B. 2π	Penny was selling tickets at the Ticket Booth. Her ticket roll shows each sale.
C. 9π	The balloon bunches were weighed one by one. All were light as feathers.
D. π	Big Lou was handing out balloons next to the Dunk Tank. A line of kids saw him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Carnival Caper

Find the area of a sector with radius 6 and central angle 60°. Leave your answer in terms of π.

A. 2π	A balloon was popped on purpose at the gate. Just a puff of air came out.
B. 36π	Big Lou was sharing a pink cloud of sugar at the Cotton Candy Stand. His sticky hands told the story.
C. 6π	The popcorn machine was opened. Just popcorn.
D. 12π	Mr. Dunk was practicing his dunking pitch near the Hall of Mirrors. A passerby recorded it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Carnival Caper

Two chords intersect inside a circle. One chord is split into pieces of length 3 and x, and the other into pieces of length 4 and 6. Find x.

A. 4.5	Ms. Taffy was refilling sugar bags and crossed through the Hall of Mirrors. The security tape shows it.
B. 7	The balloon man's bundle was checked by the fair manager. No trophy inside.
C. 24	Big Lou was waiting at the Ticket Booth for a prize ticket. Penny stamped his hand.
D. 8	Clara was painting faces at the Bumper Cars with a line of kids waiting.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Carnival Caper

A circle has circumference 18π. Find its radius.

A. 4.5	Every balloon in the park was counted and squeezed. Only balloons, no trophy.
B. 18	Big Lou was selling balloons beside the Bumper Cars all afternoon. A dozen kids bought one.
C. 36	Penny was chasing a runaway ticket stub through the Hall of Mirrors. A clerk caught it.
D. 9	The bumper cars were all inspected. Nothing was hidden in any of them.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.